Engine Model(s): PT6A

77-20-00

ENGINE INDICATING - MAINTENANCE PRACTICES

1. General

A. Maintenance personnel should make reference to the INTRODUCTION section and Chapter 70-00-00 STANDARD PRACTICES of this manual to familiarize themselves with general procedures.

B. Install suitable protective caps/covers over all disconnected tubes/lines and component openings.

C. Lockwire shall comply with specification AMS 5687, heat and corrosion resistant steel wire MS9226-03, 0.025 inch diameter, and will not be specified in instructions.

2. Consumable Materials

The consumable materials listed below are used in the following procedures.

Item No.	Name
PWC05-101	Cloth, Abrasive

3. Special Tools

Not Applicable

4. Fixtures, Equipment and Supplier Tools

The fixtures, equipment and supplier tools listed below are used in the following procedures.

Name	Remarks
Test Set	Barfield, TT1000A or TT1200A

5. Adjustment/Test

A. General

Since the bus-bar, probes and wiring harness essentially constitute a single unit, consisting of eight thermocouple probes averaged by means of an alumel/chromel bus-bar, the probes cannot be checked individually. The following tests will therefore only reveal circuit malfunctions such as an internal short circuit, or an open circuit. For individual component checks, refer to Chapter 77-20-01.

B. Operational/Functional Test

NOTE: This test is recommended to be carried out with the engine gas path component temperature close to OAT. Allow at least two hours cooling down period after engine shutdown.

(1) Remove the aircraft external leads and T1 trim thermocouple leads from the terminal block

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on the gas generator case.

(2) Connect the test set to the alumel and chromel terminals on the gas generator case terminal block and measure the loop resistance of the T5 system.

(3) Switch the test set leads to the opposite terminals on the terminal block and measure and record the loop resistance.

NOTE: The two readings must be taken in quick succession so that the temperature variation at the thermocouple level will be minimal.

(4) Add the two recorded values and divide by two to find the correct loop resistance.

(5) Measured resistance should be 0.58 to 0.74 ohm.

NOTE: If several probes are broken or damaged, the loop resistance would not necessarily fall outside the allowable tolerance. However, erroneous temperature indications could occur due to reduced sampling.

(6) To check for defective probes or harness lead, remove the engine power section assembly (Ref. 72-00-00).

(7) Connect the test set leads between either alumel or chromel terminal on the terminal block and ground (on the power section) and measure the insulation resistance. The measured insulation resistance must not be less than 50,000 ohms.

(8) Reconnect the external leads and block the trim thermocouple leads to the appropriate AL and CR terminal posts on the terminal. Tighten the bolts and torque the alumel terminal bolt 10 to 15 lb.in. and chromel terminal bolt 8 to 12 lb.in.

NOTE: Always clean all contact surfaces immediately prior to assembly with No. 400 grit abrasive cloth (PWC05-101).

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